

Interpreting Trigonometric Models

Many natural phenomena can be modeled using sinusoidal functions.

Real-world situations, such as temperature fluctuations, tidal cycles, and simple harmonic motion can be modeled using sine and cosine functions.

EXAMPLE 1

The average daily maximum temperature for Johannesburg, South Africa, can be modeled by the sinusoidal function

$$y = 8.65 \cos\left(\frac{\pi x}{6}\right) + 69.5$$

where y is the temperature ($^{\circ}\text{F}$) and x is the month ($x = 0$ corresponds to January).

A Determine the period of the function.

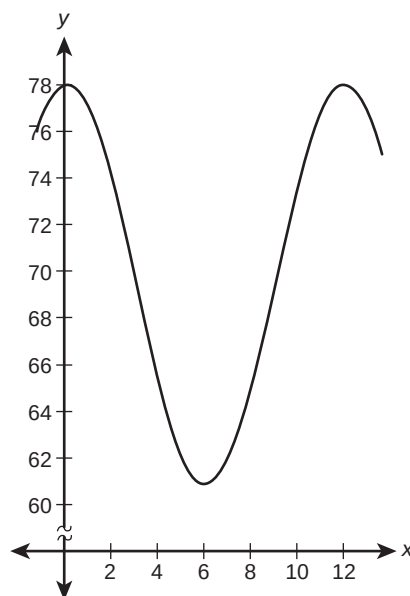
SOLUTION

Use technology to graph the function.

Since the function is in the form $f(x) = A \cos Bx + C$,

$B = \frac{\pi}{6}$ and period $= \frac{2\pi}{\frac{\pi}{6}} = 12$. So the period is 12

months, which is consistent with the graph.



► **Think About It** Because $x = 0$ corresponds to January, you have to subtract 1 from the number that is normally used to represent each month. In this model, June is 5 and December is 11.

B What is the average daily temperature during November?

SOLUTION

Substitute 10 for x in the function equation and simplify.

$$y = 8.65 \cos\left(\frac{\pi(10)}{6}\right) + 69.5 = 73.83$$

The average daily temperature during November is 73.83°F .

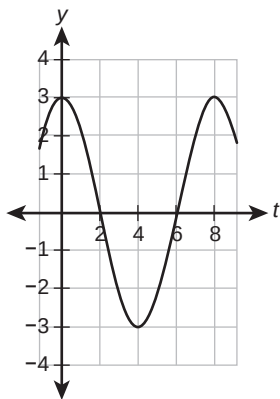
C What is the minimum average temperature, and during what month does this temperature occur?

SOLUTION

The minimum temperature can be found by subtracting the amplitude ($A = 8.65$) from the midline value (69.5), so the minimum temperature is $69.5 - 8.65 = 60.85$. You can see from the graph that the minimum temperature of 60.85° occurs during July ($x = 6$). ■

EXAMPLE 2

The movement of a weight bobbing up and down on a spring is called simple harmonic motion. The weight's vertical position over time can be modeled by a sinusoidal function. The graph shows the vertical height of the object in feet (y) as a function of time in seconds (t). Write an equation for the function.



SOLUTION

Determine which trigonometric function best models the graph.

Since a maximum occurs on the y -axis, the curve is best modeled by

$$f(t) = A \cos Bt + C.$$

► **Think About It** A graph that starts at the midline is best modeled by sine, and a graph that starts at a maximum or minimum is best modeled by cosine.

Determine the values of A , B , and C . The midline is the t -axis ($y = 0$), so $C = 0$. The maximum height reached above the midline is 3, so the amplitude, A , is 3. Also, the period is 8, so B can be found using the period formula.

$$\text{period} = \frac{2\pi}{B}$$

$$8 = \frac{2\pi}{B}$$

$$8B = 2\pi$$

$$B = \frac{\pi}{4}$$

Substitute the values for A , B , and C into the cosine function.

$$y = 3 \cos\left(\frac{\pi t}{4}\right) \quad \blacksquare$$

EXAMPLE 3

The tide chart shows the hourly tide levels (in meters) in the Bay of Fundy in St. John, New Brunswick, Canada. Create a sinusoidal model that depicts the tide level (y) as a function of time (x).

Hour	0	1	2	3	4	5	6	7	8	9	10	11	12
Tide levels (m)	4.5	3.0	1.8	1.4	1.8	3.0	4.5	6.1	7.2	7.6	7.2	6.1	4.5

SOLUTION

Determine the values of A , B , and C .

To determine C , use the midline formula.

$$\text{equation of midline} = y = \frac{\text{maximum} + \text{minimum}}{2} = \frac{7.6 + 1.4}{2} = 4.5$$

So $C = 4.5$.

To determine A , find the amplitude. Use the formula.

$$\text{amplitude} = \frac{\text{maximum} - \text{minimum}}{2} = \frac{7.6 - 1.4}{2} = 3.1$$

So $|A| = 3.1$. Also, the water level initially decreases, so $A = -3.1$.

The water starts at the midline and then after there is a minimum and a maximum level, the tide returns to the midline value after 12 h, so the period is 12.

Use the period formula to solve for B .

$$\text{period} = \frac{2\pi}{B}$$

$$12 = \frac{2\pi}{B}$$

$$12B = 2\pi$$

$$B = \frac{\pi}{6}$$

Since the midline value occurs where $x = 0$, a sine curve works best.

Substitute the values for A , B , and C into the sine function.

$$y = -3.1 \sin\left(\frac{\pi x}{6}\right) + 4.5 \quad \blacksquare$$